

Appendix A. Production-Readiness Checklist

This will be a checklist to run over all microservices—manually or in an automated way.

A Production-Ready Service Is Stable and Reliable

- It has a standardized development cycle.
- Its code is thoroughly tested through lint, unit, integration, and end-to-end testing.
- Its test, packaging, build, and release process is completely automated.
- It has a standardized deployment pipeline, containing staging, canary, and production phases.
- Its clients are known.
- Its dependencies are known, and there are backups, alternatives, fallbacks, and caching in place in case of failures.
- It has stable and reliable routing and discovery in place.

A Production-Ready Service Is Scalable and Performant

- Its qualitative and quantitative growth scales are known.
- It uses hardware resources efficiently.
- Its resource bottlenecks and requirements have been identified.
- Capacity planning is automated and performed on a scheduled basis.
- Its dependencies will scale with it.
- It will scale with its clients.
- Its traffic patterns are understood.
- Traffic can be re-routed in case of failures.
- It is written in a programming language that allows it to be scalable and performant.
- It handles and processes tasks in a performant manner.

- It handles and stores data in a scalable and performant way.

A Production-Ready Service Is Fault Tolerant and Prepared for Any Catastrophe

- It has no single point of failure.
- All failure scenarios and possible catastrophes have been identified.
- It is tested for resiliency through code testing, load testing, and chaos testing.
- Failure detection and remediation has been automated.
- There are standardized incident and outage procedures in place within the microservice development team and across the organization.

A Production-Ready Service Is Properly Monitored

- Its key metrics are identified and monitored at the host, infrastructure, and microservice levels.
- It has appropriate logging that accurately reflects the past states of the microservice.
- Its dashboards are easy to interpret and contain all key metrics.
- Its alerts are actionable and are defined by signal-providing thresholds.
- There is a dedicated on-call rotation responsible for monitoring and responding to any incidents and outages.
- There is a clear, well-defined, and standardized on-call procedure in place for handling incidents and outages.

A Production-Ready Service Is Documented and Understood

- It has comprehensive documentation.
- Its documentation is updated regularly.

- Its documentation contains a description of the microservice; an architecture diagram; contact and on-call information; links to important information; an onboarding and development guide; information about the service's request flow(s), endpoints, and dependencies; an on-call runbook; and answers to frequently asked questions.
- It is well understood at the developer, team, and organizational levels.
- It is held to a set of production-readiness standards and meets the associated requirements.
- Its architecture is reviewed and audited frequently.

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